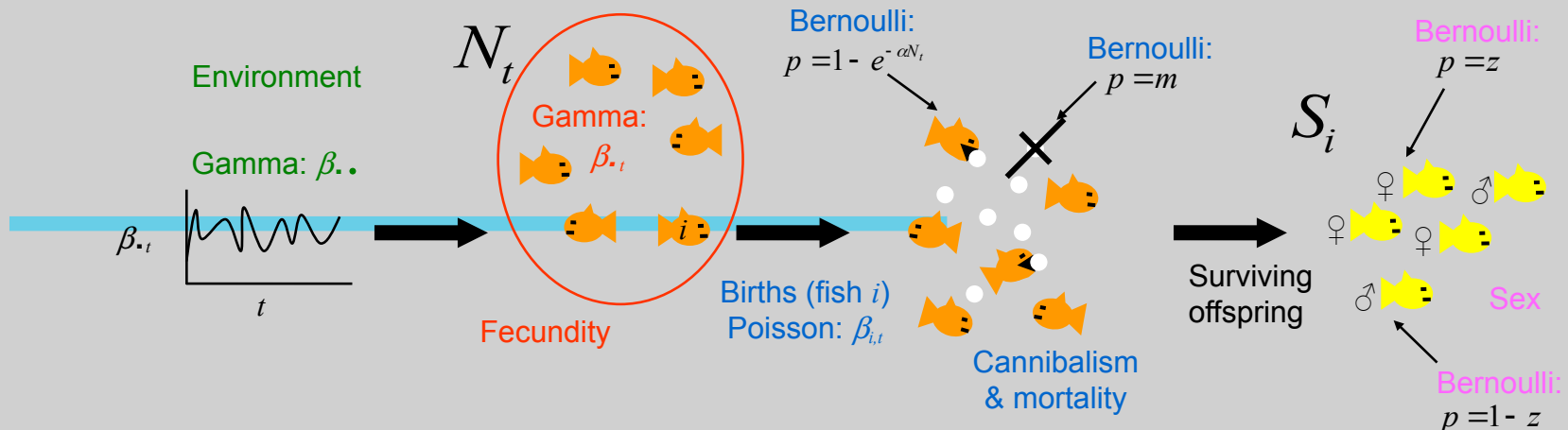


Today

- Discrete time stochastic models (cont)
- Combining distributions
- Many stochastic processes model
- Data from experiment: *Tribolium* system
- Confronting models with data: likelihood

Revisit Poisson model

- Recombining distributions
- see code



Demographic stochasticity	Environmental stochasticity	Demographic heterogeneity	Sex	Model	Abbr.
●				Poisson	P
●	●			Neg bin-environmental	NBe
●		●		Neg bin-demographic	NBd
●	●	●		Neg bin-gamma	NBg
●			●	Poisson-binomial	PB
●	●		●	Neg bin-binomial-environmental	NBBBe
●		●	●	Neg bin-binomial-demographic	NBBd
●	●	●	●	Neg bin-binomial-gamma	NBBg

All models have mean: $N_{t+1} = N_t R e^{-\alpha N_t}$

Fitting models to data

- *Tribolium* (flour beetle)
- Same life history as Ricker's fish
- Experiment:

$$N_t \rightarrow N_{t+1}$$



Likelihood

We put all the **assumptions** about the data generating process (DGP) in the **likelihood function**

I am starting with this Tribolium example because we can reasonably assume that **observation error is negligible** and all the DGP is in the ecological process

Likelihood principle

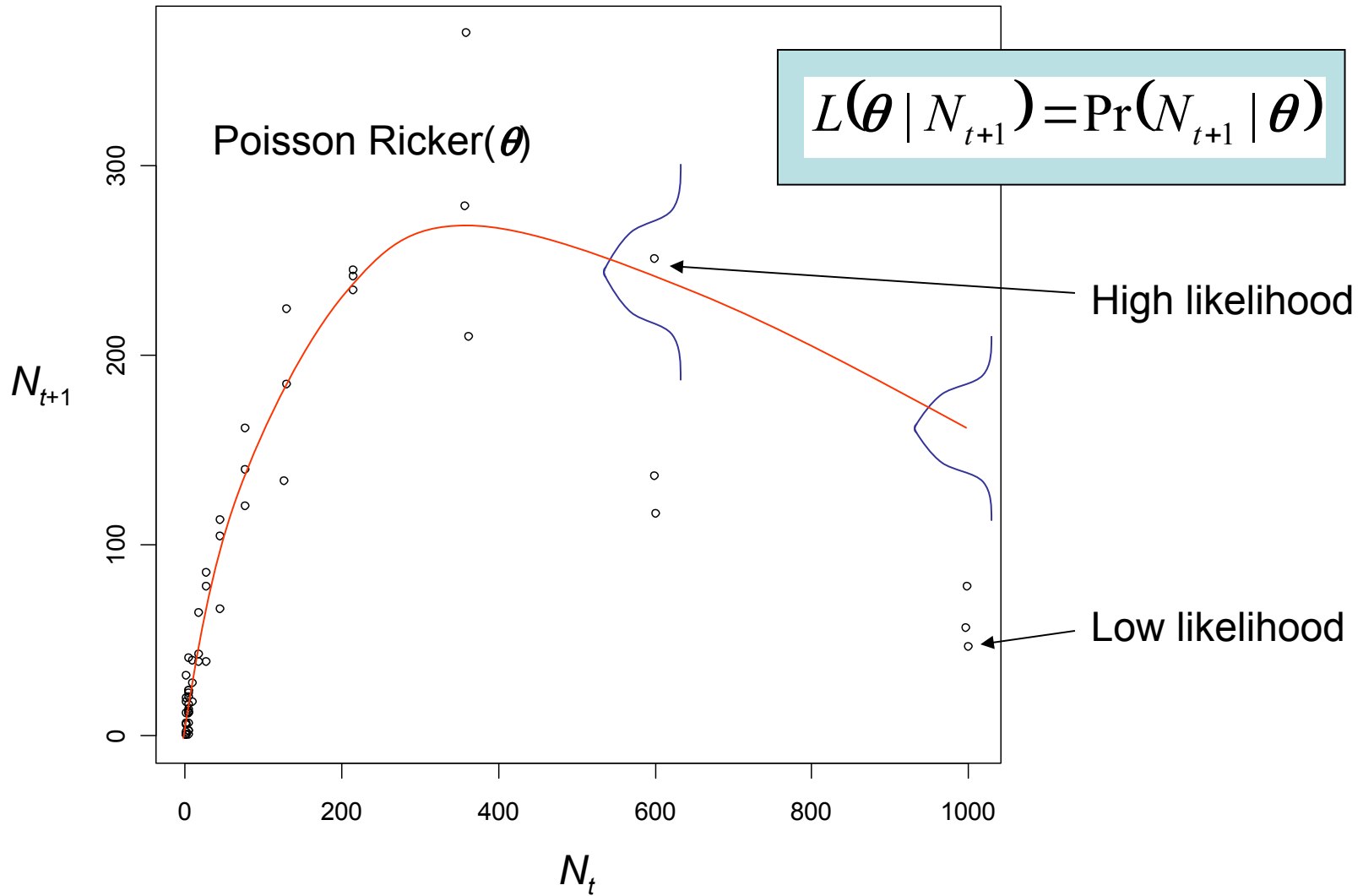
All the **evidence** in an observation (data) about the parameters (θ , model) is in the **likelihood function**

Likelihood function

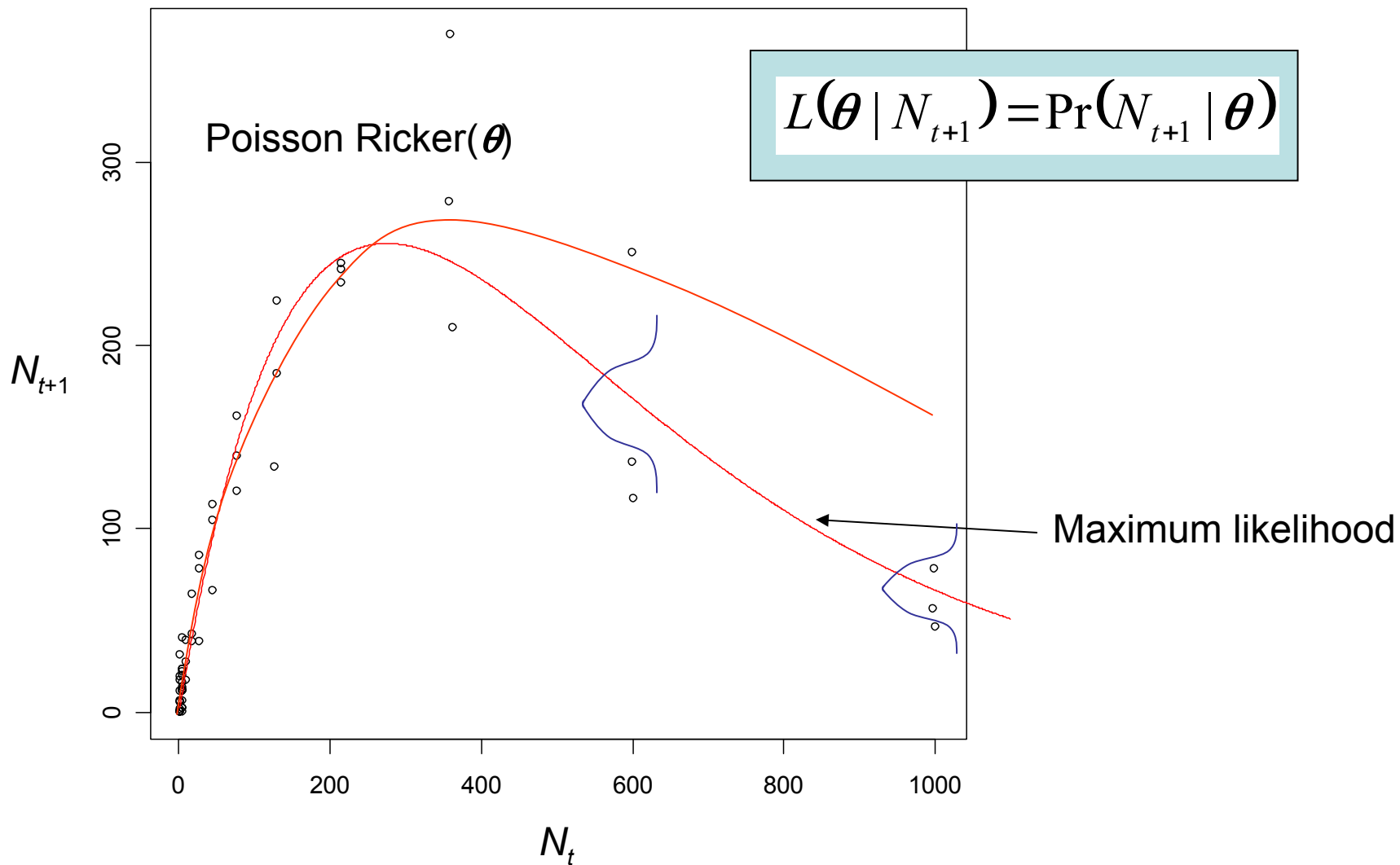
Counts all the ways the data could have happened **for a given model or hypothesis**

In this case, it's just the **PMF** of the **process model**

Likelihood



Likelihood



Training algorithm: Maximum likelihood

The **values of the parameters** that **maximize the likelihood**. In other words, the model that maximizes the probability of the data.

An **optimization** problem.

In practice: minimize the negative log likelihood. The model with the most support, has the smallest negative log likelihood.

Likelihoods

Poisson Ricker (demographic stochasticity)

$$\Pr(N_{t+1} = n_{t+1} \mid \theta, N_t = n_t) = \frac{e^{-\mu} \mu^{n_{t+1}}}{n_{t+1}!}, \quad \mu = n_t R e^{-\alpha n_t}$$

Negative binomial Ricker (environmental stochasticity)

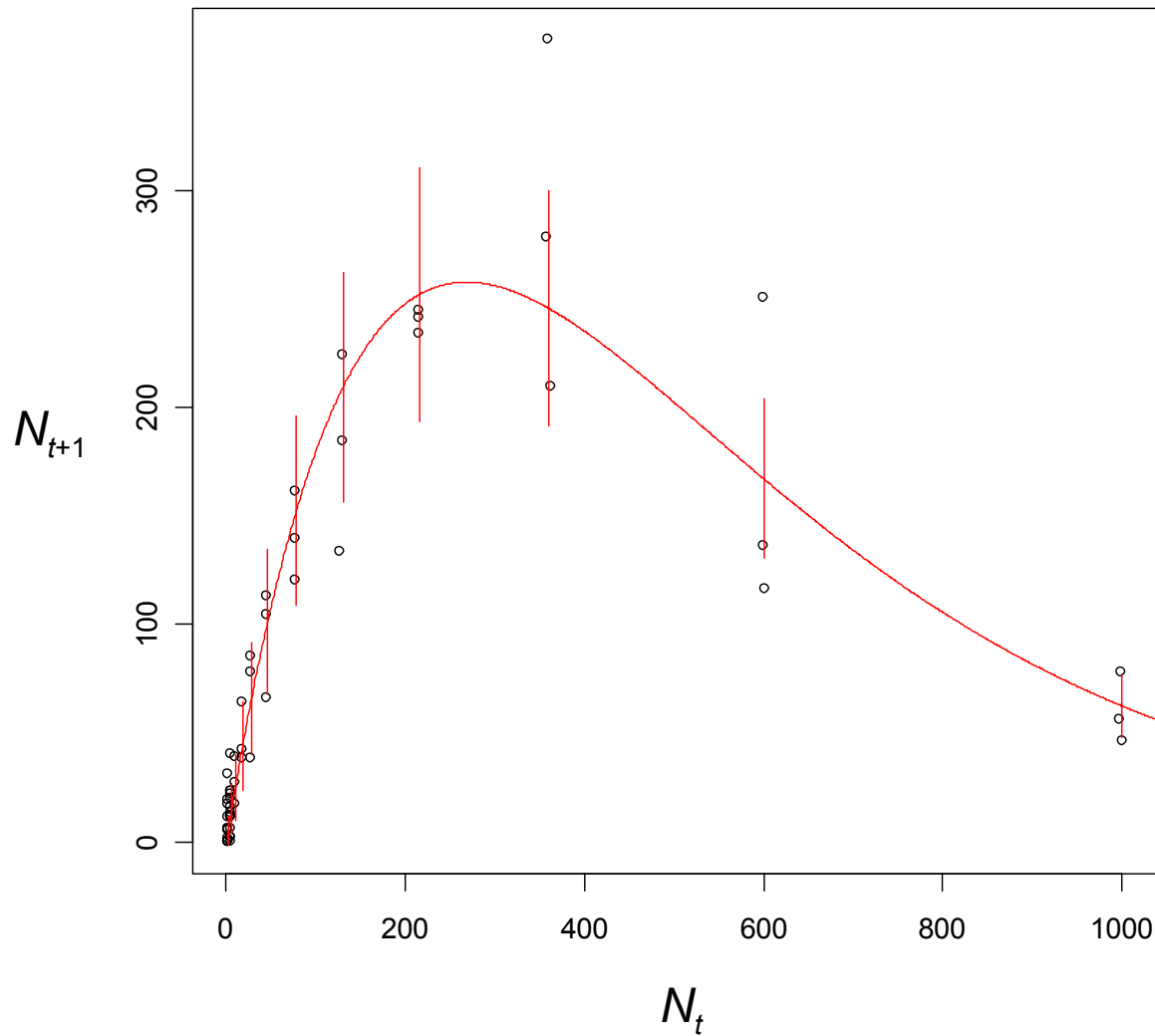
$$\binom{n_{t+1} + k_E - 1}{k_E - 1} \left(\frac{\mu}{k_E + \mu} \right)^{n_{t+1}} \left(\frac{k_E}{k_E + \mu} \right)^{k_E}$$

NBBg Ricker (all stochastic sources)

$$\int_{R_E=0}^{\infty} G(R_E) \sum_{F=0}^{n_t} \binom{n_t}{F} z^F (1-z)^{n_t-F} \binom{n_{t+1} + Fk_D - 1}{Fk_D - 1} \left(\frac{\lambda}{Fk_D + \lambda} \right)^{n_{t+1}} \left(\frac{Fk_D}{Fk_D + \lambda} \right)^{Fk_D}$$

$$G(R_E) = R_E^{k_E-1} \exp\left(-\frac{R_E k_E}{R}\right) \left(\frac{k_E}{R}\right)^{k_E} \frac{1}{\Gamma(k_E)}, \quad \lambda = F \frac{R_E}{z} e^{-\alpha n_t}$$

Best fitting model



NBBg model

All sources
of
stochasticity

Model comparison

Model	R	α	k_D	k_E	L	ΔAIC
Poisson (dem stoch)	2.526	0.003636			-406.5	336
Negative binomial (dem het)	2.638	0.003744	0.1463		-246.3	18
Negative binomial (env stoch)	2.706	0.003800		1.9913	-265.3	56
Negative binomial-gamma	2.598	0.003727	0.2610	29.2262	-238.9	5
Poisson-binomial (sex)	2.697	0.003753			-282.0	87
NB-binomial (dem het)	2.621	0.003731	0.3876		-245.8	17
NB-binomial (env stoch)	2.770	0.003831		13.1014	-242.6	10
NB-binomial-gamma	2.613	0.003731	1.1475	26.6221	-236.4	0

Demographic heterogeneity mistaken for environmental stochasticity